

- 1 (a) State Newton's second law of motion.

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[1]

- (b) A toy rocket consists of a container of water and compressed air, as shown in Fig. 1.1.

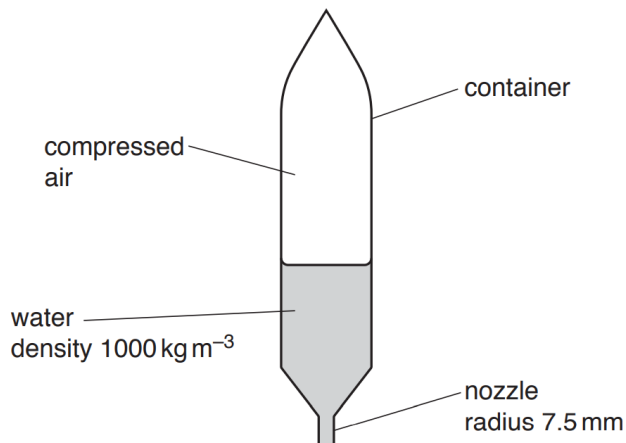


Fig. 1.1

Water is pushed vertically downwards through a nozzle by the compressed air. The rocket moves vertically upwards.

The nozzle has a circular cross-section of radius 7.5 mm. The density of the water is 1000 kg m^{-3} . Assume that the water leaving the nozzle has the shape of a cylinder of radius 7.5 mm and has a constant speed of 13 m s^{-1} relative to the rocket.

- (i) Show that the mass of water leaving the nozzle in the first 0.20 s after the rocket launch is 0.46 kg.

[2]

(ii) Calculate

1. the change in the momentum of the mass of water in **(b)(i)** due to leaving the nozzle,

change in momentum = N s [1]

2. the force exerted on this mass of water by the rocket.

force = N [2]

(iii) State and explain how Newton's third law applies to the movement of the rocket by the water.

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 [2]

(iv) The container has a mass of 0.40 kg. The initial mass of water before the rocket is launched is 0.70 kg. The mass of the compressed air in the rocket is negligible. Assume that the resistive force on the rocket due to its motion is negligible.

Calculate the acceleration of the rocket at a time of 0.20 s after launching.

acceleration = m s^{-2} [3]

[Total: 11]